

Muhammad Faisal Irshad

Associate Software Engineer | Python, FastAPI & AI Agents

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Professional Summary

Associate Software Engineer at Techanzy, working on Implement AI (implementai.io), a commercial AI Operating System with clients across the UK, Ireland, and UAE. Currently owns three production applications in parallel: **AIOS Core** (internal analytics and operations), **AIOS Portal** (no-code AI agent workflow platform), and the **Admin App** (payments, security, and access control), each built and maintained as the sole developer. Recent delivery includes a read-only MCP (Model Context Protocol) server exposing platform data to AI agents, a role-based Knowledgebase module backed by Azure Blob Storage, and a persistent workflow orchestrator with human-in-the-loop checkpoints. Strong on clean, reviewed code using DTOs, use cases, and structured architecture across FastAPI, Django, Next.js, CrewAI, and LangChain. Active GitHub contributor with 29+ public repositories.

Work Experience

Techanzy Limited

Associate Software Engineer — Implement AI

AIOS Core | Internal analytics & operations platform

Lahore, Pakistan

Feb 2025 – Present

- Built AIOS Core from the ground up as the sole developer, an internal analytics and operations platform used by the entire Implement AI team, delivering 6 of 8 planned features.
- Designed and shipped an authorized, read-only **MCP (Model Context Protocol) server** exposing platform data (clients, use cases, teams, tasks, analytics) to AI agents through a single list/view tool with no create, update, or delete path, secured with bearer-token auth and no credentials in source control.
- Registered the server as a Custom App → MCP Server integration in AIOS and validated it end to end with Claude as the test client, letting the AIOS Command team query live client records conversationally.
- Built the **Knowledgebase module**: predefined and user-defined folders with subfolders, folder-level role-based permissions, document CRUD, version history with rollback, tags, keyword search filtered by folder/tag/author/date, archive-restore, and bulk actions with partial-failure handling.
- Standardised document storage on **Azure Blob Storage** across Core and AIOS Portal, persisting file bytes in Azure with URLs and metadata tracked in the database.
- Integrated Grafana APIs and scheduled cron jobs to sync client revenue, billing metrics, agent activity, and credit usage; built a JIRA-style task management system, attendance management, RBAC across 6 user roles, and a RAG-powered support chatbot.

AIOS Portal | No-code AI agent workflow platform

- Built AIOS Portal single-handedly, giving non-technical users a simple way to configure and run AI agents from AIOS Commands, a ChatGPT-style agent product.
- Designed an n8n-style workflow builder that chains command agents into ordered steps, where each step's input is typed manually, taken from an earlier step's output, or a mix of both.
- Built the **workflow orchestrator**: owns workflow definitions, sequences steps per run, resolves each step's input, invokes the agent runtime, and tracks run and step-run status (pending, running, waiting, done, failed).
- Implemented Human-in-the-Loop (HITL) checkpoints that pause a run, surface prior output for review, and resume on submit; persisted every step's text and file artifacts to an Azure-backed artifact store.
- Built a persistent execution engine with live per-step output streaming that survives page refresh, tab close, or lost connection, plus an analytics dashboard tracking agent runs by type and credit usage.

AIOS Admin App | Payments, security & access control

- Owned and maintained the production Admin App while building Core and Portal, including access/refresh token authentication with Cloudflare Turnstile.
- Implemented Stripe Checkout and Invoice payments with webhook-driven confirmation, configured SendGrid transactional email (SPF, DKIM, domain authentication), and built the Partner Portal for introducer partners.

- Configured AI sales and support agents using CrewAI, LangChain, VAPI, and ElevenLabs, built the reusable Command Team Blueprint feature, and implemented JWT-based read-only analytics sharing links.
- Applied clean architecture (DTOs, use cases) across all three applications within a strict PR-based workflow where every change underwent team lead review.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS
Backend: FastAPI, Django, Django REST Framework, Flask, REST API Design, Clean Architecture (DTO / Use Case)
Frontend: Next.js, React, Tailwind CSS
AI & Agents: MCP (Model Context Protocol) Servers, LangChain, CrewAI, VAPI, RAG Pipelines, Workflow Orchestration, Human-in-the-Loop Design, Groq, HuggingFace, Ollama, FAISS, ElevenLabs
Data & Infrastructure: PostgreSQL, Supabase, Redis, Celery, Chroma, Azure Blob Storage, Docker, Docker Compose, GitHub Actions, Grafana, Railway, Cron Jobs
Security & Payments: JWT & Bearer Token Auth, Role-Based Access Control, Cloudflare Turnstile, Stripe Checkout & Webhooks, SendGrid (SPF/DKIM)

Projects

synQ – Job Application Tracker | *Django REST Framework, PostgreSQL, Redis, Celery, Docker, Groq* 2025

- Built a production-grade job tracking backend with JWT authentication, email verification, password reset, and TOTP-based two-factor authentication (2FA).
- Implemented async AI features (cover letter generation, job matching, interview preparation) using Celery, Redis, and Groq LLMs across modular Django apps for applications, interviews, analytics, notifications, exports, and webhooks.
- Containerized with Docker, implemented CI/CD via GitHub Actions, and wrote 143+ automated tests with pytest and pytest-django, documented through Swagger/OpenAPI.

Maternal Mental Health Monitoring System (Final Year Project) | *Flask, ML, NLP, Whisper AI, PostgreSQL* Aug 2025 –

- Developed ML models using Logistic Regression, Random Forest, and SMOTE achieving 85% accuracy on the PHQ-9 dataset.
- Built an NLP pipeline for suicide risk detection in Roman Urdu text and integrated Whisper AI speech-to-text, improving accessibility for low-literacy users.
- Developed the Flask backend in a team of three; **awarded Grade A (85%)** in FYP-1 evaluation.

Study Planner with RAG Pipeline | *Flask, FAISS, Ollama LLM, Python* Oct 2025 – Dec 2025

- Built a Retrieval-Augmented Generation pipeline over PDF, DOCX, and TXT documents, with FAISS vector search reaching 85% retrieval accuracy and context validation to reduce hallucinations.

E-Commerce Platform | *Django REST Framework, PostgreSQL, Python* Jun 2025 – Jul 2025

- Developed an e-commerce backend with 30+ REST APIs covering products, orders, checkout, and payments, on a normalized PostgreSQL schema with inventory reservation and role-based permissions.

Education

Superior University Lahore, Pakistan
Bachelor of Science in Software Engineering 2022 – 2026

Certifications & Achievements

Responsible AI Governance & AI Management System (ISO/IEC 42001:2023) – Vanguard Consulting & Inspection Service
Introduction to Python – Kaggle Certification
GitHub Portfolio – 29+ public repositories across backend, AI agent, and full-stack projects